

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Surface area

1. Find the total surface area of a solid hemisphere whose diameter is $2d$. (a) $3\pi d^2$ (b) $2\pi d^2$ (c) $\frac{1}{2}\pi d^2$ (d) $\frac{3}{4}\pi d^2$ **CBSE 2026, 1 mark**

SOLUTION

$$\text{Radius} = \frac{2d}{2} = d$$

$$\text{Total surface area of a solid hemisphere} = 3\pi r^2 = 3\pi d^2$$

Answer: (a)

2. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius r is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**

SOLUTION

A is true: the two flat circular faces fit together exactly and the curved surfaces make a complete sphere.

R is false: the surface area of a sphere is $4\pi r^2$. ($3\pi r^2$ is the total surface area of a solid hemisphere.)

Answer: (c)

3. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2 **CBSE 2024, 1 mark**

SOLUTION

$$\text{Surface area of the sphere} = 4\pi r^2$$

Each solid hemisphere has total surface area $3\pi r^2$, so the two together have $6\pi r^2$.

$$\text{Ratio} = 4\pi r^2 : 6\pi r^2 = 2 : 3$$

Answer: (c)

4. A cylinder 5 cm high has a curved surface area of 94.2 cm^2 . Find its radius, taking $\pi = 3.14$. (a) 2 cm (b) 3 cm (c) 2.9 cm (d) 6 cm **CBSE 2023, 1 mark**

SOLUTION

$$2\pi rh = 94.2: 2 \times 3.14 \times r \times 5 = 94.2, \text{ so } 31.4r = 94.2 \text{ and } r = 3 \text{ cm.}$$

Answer: (b)

5. Find the curved surface area of a cone of height 24 cm and radius 7 cm. (a) 528 cm^2 (b) 1056 cm^2 (c) 550 cm^2 (d) 500 cm^2 **CBSE 2023, 1 mark**

SOLUTION

$$\text{Slant height } l = \sqrt{24^2 + 7^2} = \sqrt{625} = 25 \text{ cm}$$

$$\text{Curved surface area} = \pi rl = \frac{22}{7} \times 7 \times 25 = 550 \text{ cm}^2$$

Answer: (c)

6. A conical cavity of height 24 cm is hollowed out of a solid cylinder of height 30 cm and radius 7 cm. The cone has the same radius as the cylinder. Find the total surface area of the solid that is left. **CBSE 2022 Term 2, 4 marks**

SOLUTION

$$\text{Slant height of the cone } l = \sqrt{24^2 + 7^2} = 25 \text{ cm}$$

The remaining solid has three surfaces: the curved surface of the cylinder, one flat circular end, and the inner curved surface of the cone.

$$\text{Total surface area} = 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l) = \frac{22}{7} \times 7 \times (60 + 7 + 25) = 22 \times 92$$

Answer: 2024 cm^2

7. A wooden article is made by scooping a hemisphere out of each end of a solid cylinder. The cylinder is 10 cm high and its base has radius 3.5 cm. Find the total surface area of the article. Take $\pi = \frac{22}{7}$. **CBSE 2018, 3 marks**

SOLUTION

The surface consists of the curved surface of the cylinder and the curved surfaces of the two hemispherical hollows.

$$\text{Total surface area} = 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$$

$$= 2 \times \frac{22}{7} \times 3.5 \times (10 + 7) = 22 \times 17$$

Answer: 374 cm^2

8. A tent is a cylinder with a cone of the same diameter on top. The cylinder is 2.1 m high and 3 m in diameter, and the slant height of the cone is 2.8 m. Canvas costs ₹500 per square metre. Find the cost of the canvas for the tent. Take $\pi = \frac{22}{7}$. **CBSE 2016, 3 marks**

SOLUTION

Radius $r = 1.5$ m, $h = 2.1$ m, $l = 2.8$ m. Canvas covers the curved surface of the cylinder and of the cone.

$$\text{Area} = 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 1.5 \times (4.2 + 2.8) = \frac{22}{7} \times 1.5 \times 7 = 33 \text{ m}^2$$

$$\text{Cost} = 33 \times 500$$

Answer: ₹16,500

9. After a flood, 50 schools agree to share equally the cost of canvas for 1500 tents. Each tent is a cylinder of base radius 2.8 m and height 3.5 m, with a cone of the same radius and height 2.1 m on top. Canvas costs ₹120 per square metre. Find the amount each school pays. Take $\pi = \frac{22}{7}$. **CBSE 2016, 4 marks**

SOLUTION

$$\text{Slant height of the cone } l = \sqrt{2.8^2 + 2.1^2} = \sqrt{7.84 + 4.41} = \sqrt{12.25} = 3.5 \text{ m}$$

$$\text{Canvas for one tent} = 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 2.8 \times (7 + 3.5) = 8.8 \times 10.5 = 92.4 \text{ m}^2$$

$$\text{Cost of one tent} = 92.4 \times 120 = 11088 \text{ rupees. Cost of 1500 tents} = 11088 \times 1500 = 16632000 \text{ rupees.}$$

$$\text{Share of each school} = \frac{16632000}{50}$$

Answer: ₹3,32,640

2 Volume

10. A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take $\pi = \frac{22}{7}$. **CBSE 2026, 3 marks**

SOLUTION

$$\text{Radius } r = \frac{7}{2} \text{ cm. Height of the cylinder} = 20 - 2r = 20 - 7 = 13 \text{ cm.}$$

$$\text{Volume of the cylinder} = \pi r^2 h = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 13 = 500.5 \text{ cm}^3$$

$$\text{The two hemispheres make one sphere: } \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \frac{539}{3} \approx 179.67 \text{ cm}^3$$

$$\text{Total volume} = 500.5 + 179.67$$

Answer: about 680.17 cm³

- 11.** A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21} \text{ m}^3$ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. **CBSE 2025, 3 marks**

SOLUTION

Let the radius be r m. Then the height of the cylinder is $2r$ m.

$$\text{Volume} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3$$

$$\frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{1408}{21}, \text{ so } \frac{176}{21}r^3 = \frac{1408}{21}$$

$$r^3 = 8, \text{ so } r = 2 \text{ and the height of the cylinder is } 2r = 4.$$

Answer: 4 m

- 12.** The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm **CBSE 2024, 1 mark**

SOLUTION

The largest cone has base diameter 2 cm and height 2 cm, so $r = 1$ cm and $h = 2$ cm.

$$\text{Volume} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 1 \times 2 = \frac{2\pi}{3} \text{ cu cm}$$

Answer: (d)

- 13.** A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm^3 . Find the two radii. **CBSE 2024, 3 marks**

SOLUTION

Let the outer and inner radii be R and r . Then $R - r = 1$.

$$\text{Volume of metal} = \pi(R^2 - r^2)h: \frac{22}{7} \times (R + r)(R - r) \times 14 = 176$$

$$44(R + r) = 176, \text{ so } R + r = 4.$$

$$\text{Solving with } R - r = 1: R = 2.5 \text{ and } r = 1.5.$$

Answer: outer radius 2.5 cm, inner radius 1.5 cm

- 14.** A room is a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21} \text{ m}^3$ of air. Find the total height of the room. Take $\pi = \frac{22}{7}$. **CBSE 2023, 3 marks**

SOLUTION

Let the radius be r m. Then the cylinder is $2r$ m high.

$$\text{Volume} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3 = \frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{176}{21}r^3$$

$$\frac{176}{21}r^3 = \frac{1408}{21}, \text{ so } r^3 = 8 \text{ and } r = 2.$$

$$\text{Total height} = \text{cylinder} + \text{dome} = 2r + r = 3r$$

Answer: 6 m

15. An ice-cream cone has radius 3 cm and height 12 cm. The lowest part of the cone, one-sixth of its volume, is left empty. The rest is filled with ice-cream, with a hemisphere of ice-cream on top. Find the volume of the ice-cream. Take $\pi = 3.14$. **CBSE 2023, 3 marks**

SOLUTION

$$\text{Volume of the cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 9 \times 12 = 36\pi \text{ cm}^3$$

$$\text{Ice-cream inside the cone} = \frac{5}{6} \times 36\pi = 30\pi \text{ cm}^3$$

$$\text{Hemisphere on top} = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \pi \times 27 = 18\pi \text{ cm}^3$$

$$\text{Total} = 48\pi = 48 \times 3.14$$

Answer: 150.72 cm³

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.